

## Week 5

# Understanding under Threat

Quantum Mechanics and Artificial Intelligence

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Hans Halvorson

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# Today's plan

## Part I: Quantum Mechanics

*(lectures 1–2)*

- The framework of QM: states, observables, dynamics
- The measurement problem and the Copenhagen myth
- Interpretations: GRW, Bohm, Everett, . . .
- Complementarity: Bohr's response to the crisis
- Bell's theorem and the limits of causal understanding

## Part II: Artificial Intelligence

*(lecture 3)*

- AI in science: prediction without understanding?
- Content to be developed from notes

From last week, quantum mechanics is our recurring example of a theory where prediction and understanding seem to come apart.

1. Do you feel that you **understand** quantum mechanics? What would it even mean to understand it?
2. Is “shut up and calculate” a satisfying attitude toward a fundamental theory of nature?

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## Two perspectives today

**First-order** Which interpretation of QM is correct? What is the theory *actually* saying about the world?

**Second-order** There has been no consensus on the foundations of QM for *100 years*. Should we be concerned? What does this tell us about science as a knowledge-producing enterprise?

Both questions bear directly on scientific understanding — which is why De Regt and Dieks chose quantum non-locality as their motivating example.

## The foundational crisis: 1900–1927

- 1900: Planck introduces the quantum of action  $h$
- 1905: Einstein explains the photoelectric effect using photons
- 1913: Bohr's model of the hydrogen atom
- 1925–26: Heisenberg's matrix mechanics; Schrödinger's wave mechanics
- 1927: Born's probability interpretation; Heisenberg's uncertainty principle; Bohr's complementarity

The early pioneers were acutely aware that they did *not* understand what was going on. Bohr thought our ability to understand the physical world was genuinely up for grabs.

## Apparent resolution: the Copenhagen school

- By 1927 a working consensus formed around Bohr's school
- Bohr's key insight: the existence of the **quantum of action** calls for a *revision* of what it means to give an objective, unambiguous description of physical phenomena
  - An unambiguous description must specify the experimental arrangement; the conditions of observation become part of the description itself
  - Classical concepts remain indispensable for describing the apparatus, but cannot all be simultaneously applied to the quantum object
- This leads to **complementarity**: a structural feature of unambiguous description, not a limitation on knowledge
- This is a positive philosophical proposal, not a counsel of despair

# Revival of the crisis

- 1952: Bohm publishes a working hidden-variable interpretation — shattering the apparent consensus
- 1957: Everett proposes the relative-state (many-worlds) interpretation
- 1964: Bell proves his non-locality theorem
- Today: active foundational debate; the cutting edge of physics proceeds without settling the foundations

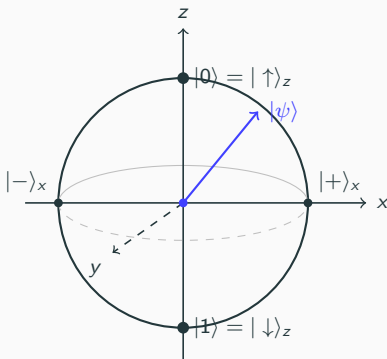
**The second-order question:** is 100 years without consensus acceptable? Does it reveal something deep about QM, or is it a sociological failure of physics?

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- The state of a quantum system is a unit vector  $|\psi\rangle$  in a complex Hilbert space  $\mathcal{H}$
- **Superposition principle:** if  $|\psi\rangle$  and  $|\phi\rangle$  are states, so is  $\alpha|\psi\rangle + \beta|\phi\rangle$  (after normalization)
- For a **qubit** (two-level system):  $\mathcal{H} = \mathbb{C}^2$
- Computational basis:  $|0\rangle \equiv |\uparrow\rangle$ ,  $|1\rangle \equiv |\downarrow\rangle$

# The Bloch sphere



Every pure state of a qubit is a point on the sphere. Antipodal points are orthogonal.

# Observables as operators

- Physical quantities: **Hermitian operators**  $A = A^\dagger$
- Possible measurement outcomes: eigenvalues  $\lambda_i$
- Eigenstates  $|a_i\rangle$ : states with definite value  $\lambda_i$

**Spin operators (Pauli matrices  $\times \hbar/2$ ):**

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

# Superpositions and the puzzle

If the system is in  $|\psi\rangle = \alpha|a_1\rangle + \beta|a_2\rangle$ , observable  $A$  has *no definite value* — yet measurement always yields a definite outcome  $\lambda_1$  or  $\lambda_2$ .

**The puzzle:** what is the system doing between preparation and measurement? Does the question even make sense?

This tension is the engine of every interpretation of QM.

# Uncertainty relations

Non-commuting observables cannot both have definite values:

$$[S_y, S_z] = i\hbar S_x \implies \Delta S_y \cdot \Delta S_z \geq \frac{\hbar}{2} |\langle S_x \rangle|$$

**Spin-z eigenstate decomposed in spin-y basis:**

$$|\uparrow\rangle_z = \frac{1}{\sqrt{2}}|\uparrow\rangle_y + \frac{1}{\sqrt{2}}|\downarrow\rangle_y$$

so  $\Delta S_y = \hbar/2$  is maximal when  $S_z$  is definite.

Conceptually: **Fourier analysis** — narrow in one basis, spread in the conjugate basis.

# Composite systems and entanglement

Composite systems:  $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ , with  
 $x \otimes (y + z) = (x \otimes y) + (x \otimes z)$ .

Not all states are product states. The **singlet state**

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

cannot be written  $|\psi\rangle_A \otimes |\phi\rangle_B$ : it is **entangled**.

**Discussion (2 min):** what features does a system have if it is entangled with another? What does it mean for its “state”?

# Linear dynamics and the Born rule

Time evolution:  $i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$ , solution  $|\psi(t)\rangle = U(t)|\psi(0)\rangle$  (**unitary**).

Crucially,  $U$  is **linear**:  $U(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha U|\psi\rangle + \beta U|\phi\rangle$ . This is exactly what generates the measurement problem.

**Born rule:**

$$\Pr_{\psi}(A = \lambda) = \langle \psi, E^A(\lambda) \psi \rangle$$

For  $|\psi\rangle = \sum_i c_i |a_i\rangle$ :  $\Pr_{\psi}(A = \lambda_i) = |c_i|^2$

**Discussion:** do we *understand* QM if it is just a recipe for computing these probabilities?

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## Definition and derivation

**Ideal measurement of  $A$ :**  $|a_i\rangle \otimes |r\rangle \longrightarrow |a_i\rangle \otimes |y_i\rangle$  (each eigenstate paired with a distinct pointer state — unitary).

Now suppose  $|\psi\rangle = \sum_i c_i |a_i\rangle$ . By **linearity**:

$$\left( \sum_i c_i |a_i\rangle \right) \otimes |r\rangle \longrightarrow \sum_i c_i (|a_i\rangle \otimes |y_i\rangle)$$

The meter ends in an **entangled superposition** — no definite reading. But we *always* observe a definite outcome.

**Contradiction.**

# The collapse postulate

**Standard “solution”:**  $\sum_i c_i |a_i\rangle \longrightarrow |a_k\rangle$  with probability  $|c_k|^2$ .

## Problems:

- Violates unitarity and linearity — the two most fundamental features of the theory
- Creates a two-dynamics theory: Schrödinger evolution *and* collapse
- When does collapse occur? Where is the “Heisenberg cut”?
- The theory gives no definition of “measurement”

# The Copenhagen myth

- Von Neumann (1932) first axiomatized collapse in *Grundlagen der Quantenmechanik*; by the 1950s it was standard textbook orthodoxy
- The label **Copenhagen interpretation** was applied retroactively — Bohr never used it
- In reality there was no single Copenhagen view: Bohr, Heisenberg, Born, Pauli all held different positions
- **HH:** Bohr never claimed the measurement problem is *solved* by collapse — his actual response is philosophically much deeper, and has been systematically misread

# Decoherence: not a solution

**Decoherence:** environmental interaction entangles the system rapidly; the *reduced* state approaches a mixture.

## Why this does not solve the measurement problem:

1. The reduced state is not *uniquely* interpretable as a mixture over definite pointer readings
2. The global state (system + environment) is still a superposition — decoherence is a local approximation, not a collapse

Decoherence explains why interference is hard to observe macroscopically. It does not explain why we observe *one* outcome.

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## Four strategies

1. **Reject unitary dynamics:** add stochastic collapse (*GRW*)
2. **Add hidden variables:** supplement QM with additional ontology (*Bohm*)
3. **Reject uniqueness of outcomes:** all branches are real (*Everett*)
4. **Revise the ontological picture:** resist the demand for a classical reality behind QM (*relational, QBism, pragmatic*)

# GRW, Bohm, and their problems

**GRW** (Ghirardi, Rimini, Weber 1986): add random spontaneous localization events. Free parameters tuned so macroscopic objects localize (no pointer superpositions) while microscopic systems remain unitary. Open: ontology? Relativistic extension?

**Bohm (1952)**: both wavefunction and particle positions are real; the wave guides particles. Particles always have definite positions.

- Guidance equation requires a preferred frame — inconsistent with special relativity
- Violates the spirit of momentum conservation in relativistic interactions

# Everett and less ontological interpretations

**Everett (1957):** unitary dynamics is complete; all branches of the wavefunction are real.

- *Probability*: deterministic theory — where do the Born-rule probabilities come from?
- *Preferred basis*: what selects the “world” branches?
- No-hidden-variables theorems imply the branches cannot themselves be classical

**Less ontological** (Rovelli, Fuchs–Schack, Healey): resist asking “what is the wavefunction in itself?”

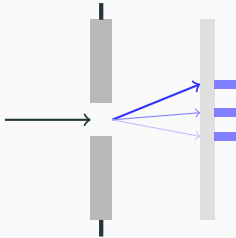
**HH:** these approaches are philosophically sophisticated but may relocate rather than dissolve the problem.

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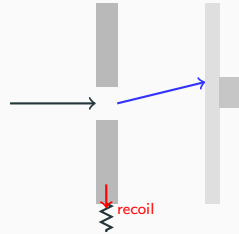
# Bohr's single-slit thought experiment

A: bolted screen



position known  $\Rightarrow$  wave picture  $\Rightarrow$   
**interference**

B: free screen



momentum known  $\Rightarrow$  particle  
picture  $\Rightarrow$  **no interference**

# Complementarity and visualizability

Two arrangements are **complementary** when they are mutually exclusive and each exhaustive within its domain.

**Spacetime description** position, trajectory, “which path” —  
**visualizable** (anskuelig) picture

**Causal description** momentum, energy, conservation laws —  
dynamical picture

“The viewpoint of complementarity presents itself as a rational generalization of the very ideal of causality.” (Bohr 1948)

This connects directly to De Regt & Dieks: Bohr is *not* saying we cannot understand quantum phenomena. He is proposing a new kind of understanding that does not require a single classical picture.

## Bohr's deeper point

- The quantum of action requires a revision of what counts as an **objective description**:
  - Objectivity does not disappear — but it now requires specifying the experimental conditions as part of any unambiguous statement about a quantum phenomenon
  - We cannot separate “how Nature is” from “the experimental arrangement in which it is observed”
- This is not about abandoning objectivity but about *reconceiving* it for a domain where the interaction between object and instrument cannot be neglected or controlled
- The misreading of Bohr as “shut up and calculate” has distorted the foundations debate for 70 years
- **HH**: Bohr is working in a tradition running from Kant through Høffding: the conditions of unambiguous description are themselves a philosophical achievement

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# EPR, von Neumann, and Kochen–Specker

**EPR (1935):** QM is incomplete — quantum correlations must be explained by predetermined hidden variables  $\lambda$ .

**Von Neumann (1932):** no hidden-variable theory can reproduce QM given additivity  $\mathbb{E}_\lambda(A + B) = \mathbb{E}_\lambda(A) + \mathbb{E}_\lambda(B)$  for non-commuting  $A, B$ . Bell (1966): this assumption is physically unmotivated.

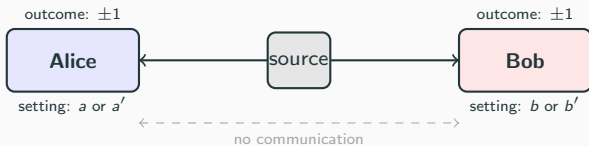
**Kochen–Specker (1967):** no assignment  $v(A) \in \text{spec}(A)$  to all observables can satisfy functional consistency  $v(f(A)) = f(v(A))$  and the sum rule for commuting pairs. **Value definiteness** is impossible.

Kochen–Specker is more compelling than von Neumann: the assumptions are operationally motivated.

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# The CHSH scenario



The entangled pair is measured far apart. Each party independently chooses a setting and records a  $\pm 1$  outcome.

## Local hidden variables and the CHSH inequality

Suppose outcomes are determined by  $\lambda \sim \rho(\lambda)$ , with **reality** ( $A, B$  definite) and **locality** (Alice independent of Bob's setting):

$$E(a, b) = \int A(a, \lambda) B(b, \lambda) \rho(\lambda) d\lambda$$

Under local hidden variables:

$$|E(a, b) - E(a, b') + E(a', b) + E(a', b')| \leq 2$$

*Proof sketch:* for each  $\lambda$ , the combination  $A(a)[B(b) - B(b')] + A(a')[B(b) + B(b')]$  takes values in  $\{-2, 0, +2\}$ .

# Quantum violation

For  $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ :  $E(a, b) = -\cos(\theta_a - \theta_b)$

Choose  $a = 0^\circ$ ,  $a' = 90^\circ$ ,  $b = 45^\circ$ ,  $b' = 135^\circ$ :

$$E(a, b) = -\cos(-45^\circ) = -\frac{1}{\sqrt{2}}$$

$$E(a, b') = -\cos(-135^\circ) = +\frac{1}{\sqrt{2}}$$

$$E(a', b) = -\cos(45^\circ) = -\frac{1}{\sqrt{2}}$$

$$E(a', b') = -\cos(-45^\circ) = -\frac{1}{\sqrt{2}}$$

$$S = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{4}{\sqrt{2}} = -2\sqrt{2} \quad \Rightarrow$$

$$|S| = 2\sqrt{2} \approx 2.83$$

# The Jarrett analysis and what Bell means

Jarrett (1984): local hidden variables = two conditions:

**PI (parameter independence)** Alice's outcome does not depend on Bob's *setting*

**OI (outcome independence)** Alice's outcome does not depend on Bob's *outcome*

Bell violation  $\Leftrightarrow$  at least one of PI or OI fails. Bohm/Everett: OI fails. Retrocausal models: PI fails.

**HH's focus:** Bell's theorem shows a fundamental limitation on **local causal explanation** of quantum correlations. Does this mean we *cannot understand* them?

De Regt & Dieks: understanding does not require causal explanation.  
Bohr: we need a new conceptual framework, not a new causal story.

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- QM works to extraordinary precision (electron  $g$ -factor: 12 significant figures)
- Yet the theory is arguably *inconsistent as stated* (the measurement problem), and there is no consensus on how to resolve it
- Each interpretation pays a significant price: non-locality, many worlds, anti-realism, new fundamental dynamics
- **HH:** this is a genuine crisis — not merely a philosophical puzzle we can comfortably set aside

- Some physicists say: “QM works, so who cares if we don’t understand it?”
- If we accept this for QM, we have weakened our ground against the parallel argument:  
*“AlphaFold predicts protein structures better than any human theory — so who cares if we don’t understand it?”*
- **HH:** I do not think we should be content with this. The goal of science is not mere prediction but understanding. QM is one thing; the threat from AI is another.

## **Part II**

Artificial Intelligence and Scientific Understanding

## 9. AI and Scientific Understanding

*Notes and content for this section will be added.*

